

Alfido Tech Internship

Task 1: Temperature Monitoring & LCD Display

Name:

Praveena Reddy

Domain:

Embedded Systems

Tools Used:

- Tinkercad Circuit Simulator
 - Arduino IDE
-

1. Objective

The objective of this project is to monitor temperature using an temperature sensorTMP36 and display the temperature value on a 16x2 LCD display. The measured temperature is also logged through the Serial Monitor for monitoring purposes.

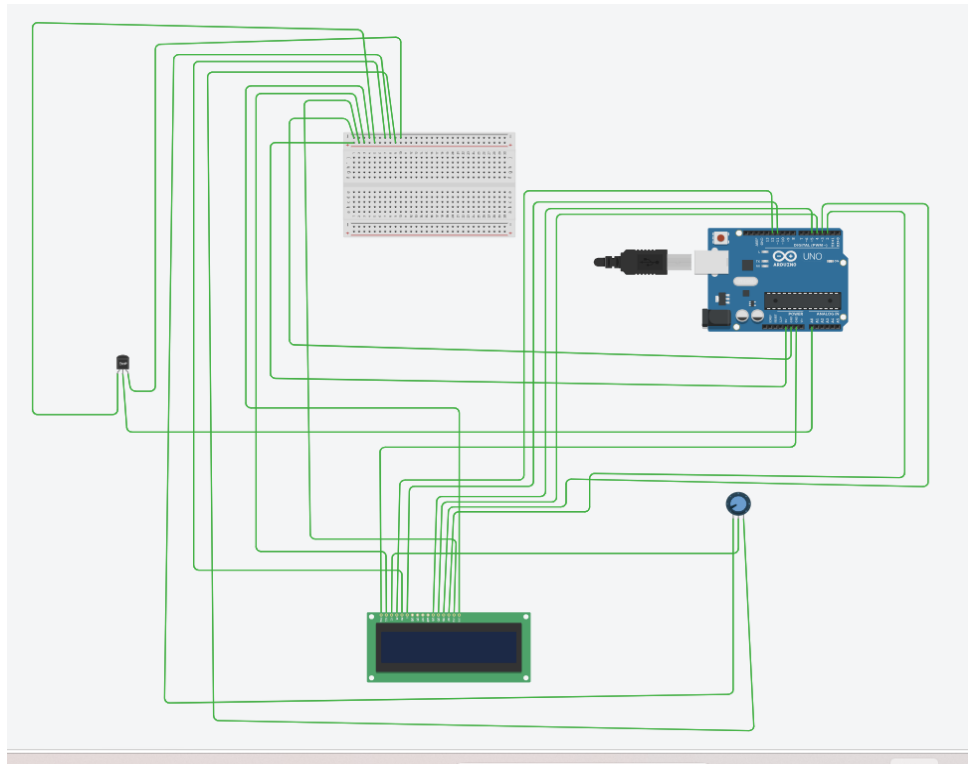
2. Components Used

- Arduino UNO
 - Temperature Sensor TMP36
 - 16x2 LCD Display
 - Breadboard
 - Jumper Wires
-

3. Software Used

- Tinkercad Circuit Simulator
 - Arduino IDE
-

4. Circuit Diagram



5. Working Principle

The TMP36 temperature sensor measures the surrounding temperature and produces an analog voltage output that varies according to the temperature. The Arduino UNO reads this analog voltage through its analog input pin and converts it into temperature values in degrees Celsius. The calculated temperature is displayed on the 16x2 LCD display and simultaneously logged on the Serial Monitor. This enables real-time temperature monitoring and display.

6. Arduino Code

```
#include <LiquidCrystal.h>
```

```
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
```

```
int sensorPin = A0;
```

```
void setup()
```

```
{
```

```
  lcd.begin(16, 2);
```

```
Serial.begin(9600);

lcd.print("Temp Monitor");
delay(2000);
lcd.clear();
}

void loop()
{
  int reading = analogRead(sensorPin);

  float voltage = reading * (5.0 / 1024.0);

  float temperatureC = (voltage - 0.5) * 100.0;

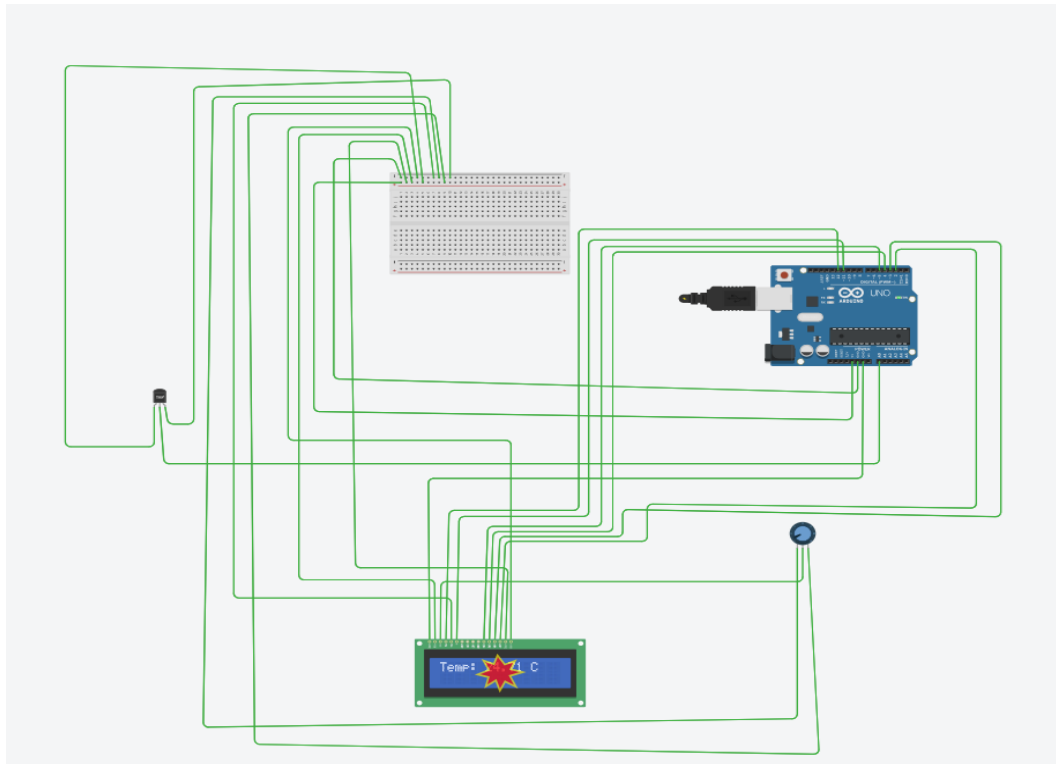
  lcd.setCursor(0,0);
  lcd.print("Temp: ");
  lcd.print(temperatureC);
  lcd.print(" C");

  Serial.print("Temperature: ");
  Serial.print(temperatureC);
  Serial.println(" C");

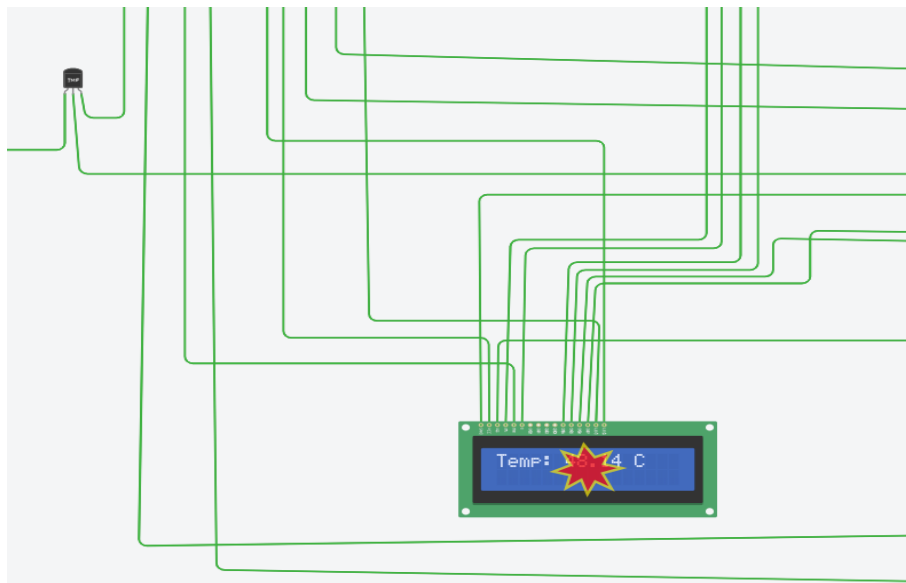
  delay(1000);
}
```

7. Output Screenshots

Circuit Screenshot



LCD Display Output



Serial Monitor Output

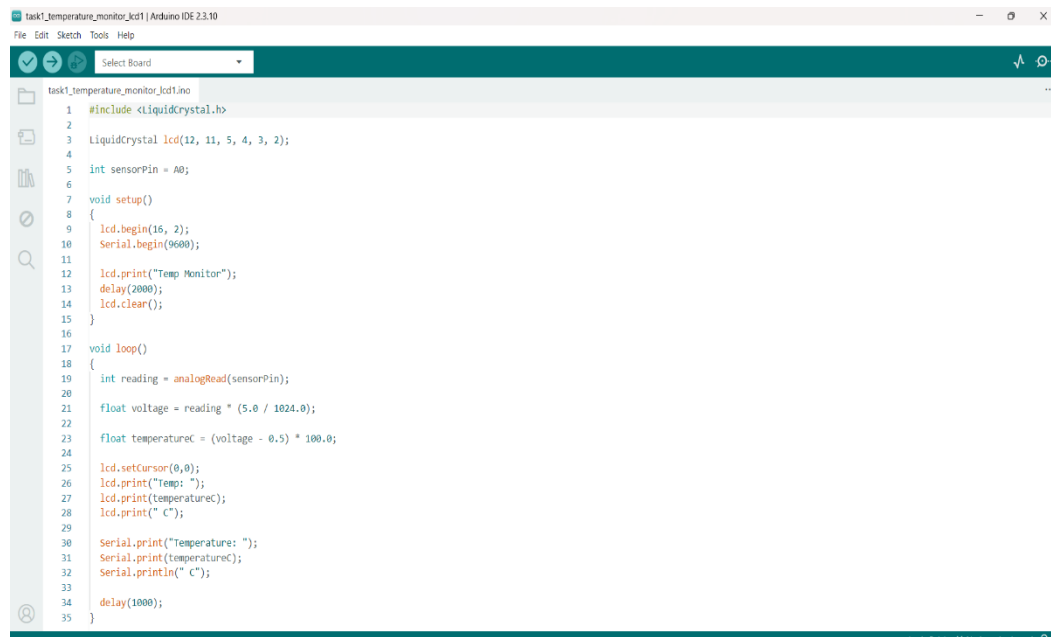
Serial Monitor

```

Temperature: 24.71 C
Temperature: 24.71 C
Temperature: 24.71 C
Temperature: 48.14 C
Temperature: 48.14 C
Temperature: 48.14 C
Temperature: 48.14 C

```

Code Screenshot



```
task1_temperature_monitor_lcd1.ino
1 #include <LiquidCrystal.h>
2
3 LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
4
5 int sensorPin = A0;
6
7 void setup()
8 {
9   lcd.begin(16, 2);
10  Serial.begin(9600);
11
12  lcd.print("Temp Monitor");
13  delay(2000);
14  lcd.clear();
15 }
16
17 void loop()
18 {
19   int reading = analogRead(sensorPin);
20
21   float voltage = reading * (5.0 / 1024.0);
22
23   float temperatureC = (voltage - 0.5) * 100.0;
24
25   lcd.setCursor(0,0);
26   lcd.print("Temp: ");
27   lcd.print(temperatureC);
28   lcd.print(" C");
29
30   Serial.print("Temperature: ");
31   Serial.print(temperatureC);
32   Serial.println(" C");
33
34   delay(1000);
35 }
```

8. Results

The temperature monitoring system successfully measured the surrounding temperature using the TMP36 temperature sensor. The temperature readings were accurately displayed on the 16x2 LCD display and simultaneously logged through the Serial Monitor. The project was tested in the Tinkercad simulation environment and functioned as expected, providing real-time temperature monitoring.

9. Conclusion

This project demonstrates the basic concepts of embedded systems including sensor interfacing, analog data acquisition, LCD interfacing, and serial communication using Arduino. The system successfully monitored and displayed temperature readings in real time.

10. GitHub Repository Link

[reddypraveena239-prog/Task1-Temperature-Monitoring: Task 1 - Temperature Monitoring and LCD Display using Arduino UNO and TMP36 Sensor](https://github.com/reddypraveena239-prog/Task1-Temperature-Monitoring: Task 1 - Temperature Monitoring and LCD Display using Arduino UNO and TMP36 Sensor)

11. Demo Video Link

https://drive.google.com/file/d/1A9rKQ_FZLDCGDdeFFyUXs8c7Q3LNh3Qh/view?usp=sharing